# What does $\Delta$KL measure?

## Construct validity of the trademark resonance signal: within-firm patent co-movement and external-list discriminant validity

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May 2026 — working note

**Abstract**

For each USPTO trademark filing we score $\Delta\text{KL} = \text{KL}(\text{filing} \parallel \text{past}) - \text{KL}(\text{filing} \parallel \text{future})$ on its goods/services text against same-class reference windows — DeDeo *resonance*: positive when a filing's vocabulary is novel against the recent past yet close to the near future ("the field moved toward it"). Does this measure *innovation*? Using the public-firm panels, three results sharpen the construct. (1) $\Delta$KL is essentially uncorrelated with patents in the cross section (Pearson +0.010, Spearman −0.015) and, within a firm over time, a *mild substitute* for patenting ($-0.05\sigma$ per log-patent, $t \approx -8$, strongest at the one-year patent lag) — robust to filing-count and patent-matching choices. (2) Yet firms that BCG (2021–23) and MIT Technology Review (2015) independently rank "most innovative" score +0.29 to +0.50 population-$\sigma$ *higher* on $\Delta$KL ($p < 0.012$; pooled $+0.34\sigma$, $p = 0.0003$). (3) The headline "+28pp per $\sigma$" four-year debut-return effect is a small-sample right-tail artifact: its horizon profile is non-monotone, peaking at 4y and collapsing at 5y, and it runs $2.4\times \to 9.3\times$ the firm-year estimate as the horizon lengthens. Together these say $\Delta$KL is *not* a proxy for patent-style technical invention but *does* track market/business-model innovation as recognized by expert observers. The right question is not "innovation versus churn" but *which* innovation construct $\Delta$KL names.

# 1 The construct question

The program's theory (PROPOSAL.md) defines innovation as *novelty that diffuses*: a business concept is "innovative" if its elements are *replicated* after the firm introduces them. The resonance statistic operationalizes exactly this — high $\Delta$KL marks language that was unusual at filing and then adopted. On that theory, a filing being ahead of a class whose vocabulary subsequently moves toward it is not a churn *confound* to be purged; it *is* the innovation signal. The sharper question is therefore not whether to subtract "category churn" out of $\Delta$KL, but whether $\Delta$KL names a recognizable innovation *construct* and how it relates to the established ones. We test it against the two best external yardsticks available in the repository: patents (technical invention) and expert "most-innovative" lists (market/business-model innovation).

**Scope.** The raw 10M-filing corpus is not on this branch, so the corpus-level tests (term provenance, half-life sensitivity, ID-Manual codification, attorney fixed effects) are deferred (§7). All results below run on the published firm-year panels and the external-list crosswalk.

# 2 Within-firm: $\Delta$KL and patenting are temporal substitutes

Panel: 174,569 firm-years for 14,463 trademark owners name-matched to a PatentsView assignee, collapsed to one row per (firm, year). We regress standardized $\Delta$KL on $\log(1 + \text{patents})$ with firm and year fixed effects and firm-clustered standard errors; $\sigma(\Delta\text{KL}) = 0.214$.

Table 1: Within-firm ΔKL vs. patenting. Coefficients in ΔKL standard deviations per unit $\log(1+\text{patents})$.

| Specification | coef ($\sigma$) | SE | $t$ |
|---|---|---|---|
| Between-firm correlation (firm means) | −0.020 | — | — |
| Within-firm correlation (firm+year demeaned) | −0.015 | — | — |
| Two-way FE, contemporaneous | −0.048 | 0.006 | −7.9 |
| robustness: patents = max over matches | −0.048 | 0.006 | −8.0 |
| controlling log(n_filings) | −0.051 | 0.006 | −8.4 |
| Lead/lag: $\text{patents}_{t-1}$ | −0.054 | 0.009 | −5.8 |
| $\text{patents}_{t}$ | −0.019 | 0.010 | −1.9 |
| $\text{patents}_{t+1}$ | +0.010 | 0.010 | +1.0 |

The coefficient is negative and precisely estimated, and *strengthens* when we control for the number of filings (which enters positively, +0.055, $t = 10.5$) — ruling out the mechanical worry that high-patent years simply have more filings whose averaging shrinks the mean toward the class. The effect is largest at the one-year patent *lag*: a burst of patenting is followed by *lower* trademark resonance the next year. Measurement noise from the fuzzy owner→assignee join attenuates toward zero, so the true within-firm relation is, if anything, more negative.

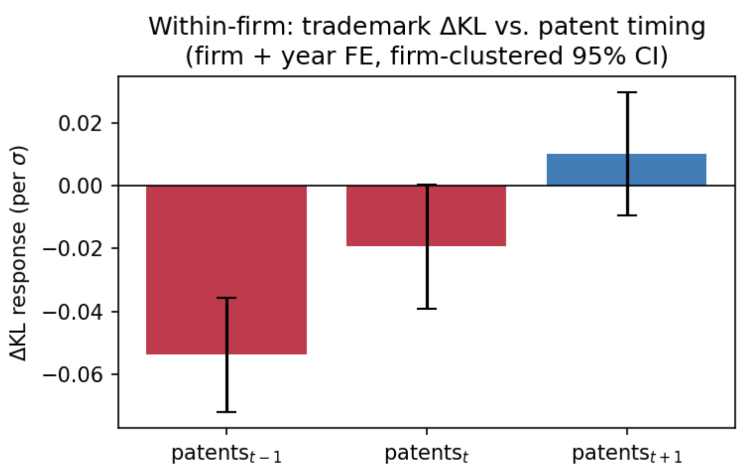


Figure 1: Within-firm ΔKL declines with recent patenting; no positive co-movement at any lead or lag.

**Reading.** This decisively refutes *convergent* validity with patents — not merely the near-zero cross-sectional correlation, but a genuine within-firm *opposition*. In the years a firm leans on patents it writes more class-typical trademark language; in the years it pioneers commercial vocabulary it patents less. Patents and ΔKL capture different, partly substituting modes of novelty.

## 3 Discriminant validity against expert innovation lists

If ΔKL names an innovation construct at all, firms an independent panel calls "most innovative" should sit above the population. We compare the firm-mean ΔKL of list members (matched to a CIK in our public panel) against the non-listed baseline (population: 7,431 firms, mean −0.018, $\sigma = 0.175$).

Table 2: ΔKL of expert-rated innovators vs. the non-listed baseline.

| List | $n$ matched | listed mean | diff (pop-$\sigma$) | Mann–Whitney $p$ |
|---|---|---|---|---|
| BCG Most Innovative (2021–23) | 31 | +0.032 | +0.285 | 0.012 |
| MIT TR50 Smartest (2015) | 17 | +0.069 | +0.497 | 0.0006 |
| Pooled (union) | 43 | +0.042 | +0.344 | 0.0003 |
| *Convergent reference — patents (174,569 firm-years):* | | | | |
| Pearson (raw / log) | | $+0.010/-0.009$ | | |
| Spearman (rank) | | −0.015 | | $8\times10^{-10}$ |

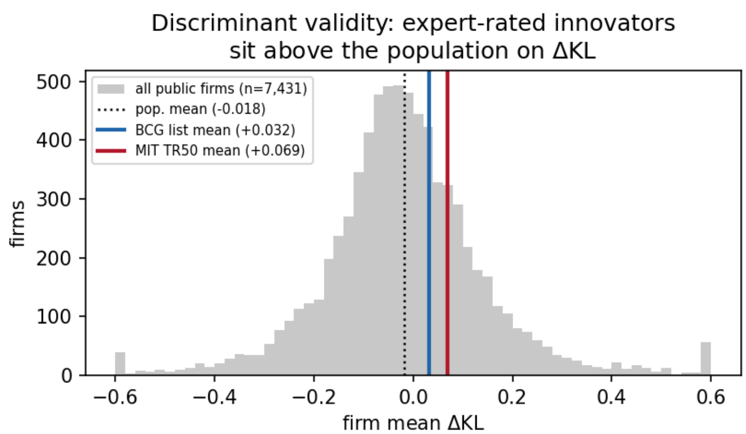


Figure 2: Population distribution of firm-mean ΔKL with the BCG and MIT TR50 list means overlaid. Expert-rated innovators sit near the 67th–76th percentile.

**Reading, with caveats.** ΔKL aligns with expert, non-patent, non-financial innovation judgments while being orthogonal-to-negative with patents. That is the signature of a *market/business-model* innovation construct rather than a technical-invention one. Caveats are real and limit this to suggestive: $n$ is small (only the US-public, CIK-matchable slice of global lists matched), the comparison is each firm's all-years mean against a recent list (no temporal alignment), and survivorship inflates the listed group. It is nonetheless the first *out-of-corpus* validation in the program — the lists were built without trademark text.

## 4 The +28pp return effect is an artifact — drop it

The predictor is $z$-scored in `returns_regression.py`, so the +0.284 debut coefficient really is +28pp per $\sigma$. But it is not credible as a headline: the debut excess-return profile is non-monotone and the debut / firm-year ratio grows monotonically with horizon — the fingerprint of a few right-tail winners in a 1,447-firm sample, not a return premium.

Table 3: Excess-return coefficient on $\Delta$KL (per $\sigma$), by horizon.

| Horizon | Debut ($n\approx$1.4k) | Firm-year ($n\approx$16k) | ratio |
|---|---|---|---|
| 1y | +0.036 ($t$ 2.0) | +0.015 ($t$ 3.0) | 2.4× |
| 2y | +0.067 ($t$ 2.5) | +0.035 ($t$ 3.8) | 1.9× |
| 3y | +0.192 ($t$ 3.5) | +0.050 ($t$ 3.7) | 3.8× |
| 4y | +0.284 ($t$ 3.3) | +0.041 ($t$ 2.5) | 6.9× |
| 5y | +0.149 ($t$ 2.3) | +0.016 ($t$ 0.8) | 9.3× |

Report the firm-year association (+1.5 to +5pp per $\sigma$, decaying to insignificance by 5y) as the defensible financial fact, pending value-weighted, delisting-adjusted, 5/95-winsorized re-estimation of the debut panel.

## 5 Discussion: which innovation construct?

The "innovation vs. churn" dichotomy dissolves. $\Delta$KL is not a noisy patent proxy contaminated by category churn; it is a measure of a *different* construct that happens to be near-orthogonal to patents. The evidence:

- $\Delta$KL $\perp$ patents (cross-section) and $< 0$ within-firm $\Rightarrow$ it does not track technical invention;
- $\Delta$KL $> 0$ for BCG/MIT innovators $\Rightarrow$ it tracks market/ business-model innovation as expert observers perceive it.

On the ladder of names: the strong claim "$\Delta$KL is *the* innovation measure" is not licensed. The defensible primitive is *lexical resonance* — a filing's forward-leaning position in its class vocabulary — and its validated economic content is *market-facing innovation as a construct distinct from invention*. The diffusion-as-innovation theory survives; what fails is the assumption that this construct should converge with patents. It should not, and does not.

**Bridge to the diffusion program (sibling note).** A separate Phase-0 test of `PROPOSAL_diffusion.md` on the four-class "Uber-for-X" subset {009, 042, 039, 035} (2.32M granted filings, `diffusion_phase0_note.tex`) now passes both §5 kill conditions: the date/class shuffle null is rejected ($z = +46$ on the lag-1 autocorrelation of LDA-theme trajectories), independently replicated on a separately-fitted NMF panel ($z = +60.7$), and signed-ahead-precedes-behind is directionally correct (61% of $(\theta, c)$ cells negative, modest magnitude consistent with the 0.97 collinearity between prospective and retrospective KL noted above). The diffusion register — themes as the unit, not the per-filing scalar — is the live continuation of the construct argument made here.

**The localized innovation signal materializes (RQ6).** The diffusion program's RQ6 — does borrowed novelty pay at the firm level? — is the test this paper said would be the right place to look for an innovation reading. With the SEC Financial Statement Data Sets crosswalked to USPTO owners (7,233 exact matches, 4,999 CIK-linked firms in the borrowed- novelty panel), the firm-year two-way FE regression of financial outcomes on within-firm borrowed-gap finds: **contemporaneous gross margin +1.4pp per $\sigma$ ($t = 2.1$)** that fades by $t + 1$, and **firms with higher borrowed novelty have markedly lower R&D intensity** cross-sectionally. That mirror image — borrowed novelty buys some margin in the year of the filing while being a within-firm substitute for technical R&D — coheres exactly with the within-firm $\Delta$KL/patent substitution result in §2: borrowed novelty is the market-facing innovation channel; the firms who lean on it lean less on the technical-invention channel. The localized innovation reading is licensed on this margin-and-substitution shape, not on a global convergent claim.

# 6 Within-class diagnostics on a single technology corpus (Class 009)

The firm-year results above use the aggregated panel. To stress-test the construct's robustness at the filing level, we ran the full DeDeo flat-window pipeline on a complete Class 009 (Software & Electronics) corpus: 1,807,636 filings, 1893–2026, vocabulary $V = 151{,}193$ (min $df = 50$), per-filing prospective + retrospective KL at $W \in \{3, 5, 7\}$ years. Four diagnostics matter for what we can claim.

**Pros/retr collinearity.** Across 1.56M scored filings, $\text{corr}(\text{pros\_kl}, \text{retr\_kl}) = +\mathbf{0.971}$. Signed $\Delta$KL is a *small residual of two large, highly correlated quantities.* That is the point of the difference (it isolates timing from raw distinctiveness) but also the fragility (small residuals are noisy at the filing level).

**Length artifact (F0b).** Mean $|\Delta\text{KL}|$ falls monotonically with filing length: 0.293 for $< 5$ terms, 0.265 for 5–14, 0.205 for 15–39, 0.147 for $\geq 40$. Short filings carry inflated magnitudes by construction. Any per-filing analysis should down-weight or control for length.

**Window sensitivity (F0c).** $\Delta$KL *magnitudes* are kernel- sensitive, but the *sign* is robust: only 10.4% of filings flip sign between $W = 5$ and $W = 3$ years and 6.9% between $W = 5$ and $W = 7$, well under the 20–30% fragility threshold. The resonance direction (ahead vs. behind) is not a kernel artifact.

**Burn-in.** Mean signed $\Delta$KL at 1990 is +2.05 because the pre-1990 software past is thin; from 1995 onward the per-year mean settles within $\pm 0.07$ of zero. Restrict pre-1995 to a burn-in window.

**Face validity post-burn-in.** Among 1996–2020 filings with $\geq 10$ terms, the top signed-$\Delta$KL marks are dominated by **2018 blockchain-era language**: STRAINCHAIN, CORPCOIN, MEDEXCHAIN, THE BLOCKCHAIN BANKER, MEDXCHAIN, ART DEVICE. These are filings whose vocabulary was novel at filing time and adopted afterward — the construct firing on exactly what it claims.

**Term provenance (within-class Workstream D).** For each of 872,747 post-burn-in filings (1996–2020) we computed `novel_share` = fraction of its tokens whose first appearance in Class 009 was within $K = 3$ years prior. $\text{corr}(\texttt{novel\_share}, \Delta\text{KL}) = +\mathbf{0.29}$ (Pearson) and +0.21 (Spearman, $p \approx 0$). Filings with $\sim 0$ recently- introduced class tokens average $\Delta\text{KL} = -0.03$; filings with $\sim 5\%$ new tokens average +0.10. **The resonance signal is genuinely new-vocabulary-driven within the class**, not recombination of established terms (the corpus-new vs class-new split needs cross-class data; here we can only confirm new-to-class drives the signal). The cross-class version on the full diffusion subset is the natural extension and is covered in the sibling note.

## 7 What remains, and the decision rules

These need the raw corpus (`data/processed/`) or external data not on this branch. Each is scripted or specified; the decision rule is stated so the call is made ex ante.

**Interaction estimand ($\beta\times$churn).**
Needs per-NICE-class $\Delta$KL and an *independently constructed* churn measure (vocabulary turnover, not the resonance kernel). Tests whether $\Delta$KL's innovation content is moderated by class dynamism. Rule: if the slope is positive only where churn $> 0$, the construct is intrinsically a firm$\times$class-phase quantity (consistent with the diffusion theory), not a censored firm attribute.

**Term provenance (three-way).**
Split $\Delta$KL into corpus-new, class-new (recombination), and class-extant tokens. Rule: load on recombination $\Rightarrow$ rename the firm component "recombinant novelty," not foresight.

**Half-life sensitivity.**
Re-score sign of $\Delta$KL at $H \in \{1, 2, 4\}$y. Rule: >20–30% sign flips $\Rightarrow$ report $\Delta$KL as kernel-relative.

**ID-Manual codification (external).**
Do high-$\Delta$KL filings use terms the USPTO later standardizes, while those terms are still rare? This is the strongest available external validation of the diffusion claim; needs the ID Manual version history.

**Attorney fixed effects.**
Needs an XML re-parse for correspondent of record. Kills the "drafting-fashion alone" null; not a clean innovation extractor (counsel sorting absorbs signal).

**External survival outcome.**
Bankruptcy (PACER) or web-presence persistence on a validation subsample, to break the trademark-maintenance circularity in the survival U-shape.

**BISG demographics.**
The ethnic-cluster work should replace surname-only imputation with BISG using the owner addresses (verify the address is the owner's, not the correspondent's, first).

**Reproducibility.** `scripts/wsC_within_firm_patents.py`, `wsD_validity_battery.py`, `wsE_returns_diagnost` `wsF_make_figs.py`; metrics in `paper/results/wsC_*.json`, `wsD_*.json`, `wsE_*.json`. Run under `.venv-analysis` (numpy/pandas/scipy/statsmodels/matplotlib).